Practical Five
The Identification and interpretation of bedforms and sedimentary structures in the rock record

> "*As a young man, my fondest dream was to become a geographer. However while working in the customs office I thought deeply about the matter and concluded it was far too difficult a subject. With some reluctance, I then turned to Physics as a substitute*."
>
> - Albert Einstein (Unpublished Letters)

Bedforms and their sedimentary structures are the building blocks of the stratigraphic record. The ability to recognise the sedimentary structures and their formative bedforms is fundamental to the interpretation of depositional environments.

Intended Learning outcomes
At the end of this practical class you should be able to:

- identify the basic families of primary and secondary bedforms and structures commonly found in the rock record
- draw accurate, annotated sketches, such that the bedform/structure is identifiable by a third party from your diagram
- identify which fabrics, bedforms and sedimentary structures can be used to determine the way up of the stratigraphy, and how to do it
- identify which fabrics, bedforms and sedimentary structures can be used to collect palaeocurrent data, and how to do it

Part 1 The identification of common bedforms and sedimentary structures
Around the room are specimens of common bedforms and sedimentary structures.

Create a classification scheme for organising the bedforms and structures into related groups. Think carefully about the basis of your classification scheme.

For each fabric, bedform or sedimentary structure, specify

a. draw a careful annotated sketch to emphasise the key observations used to identify the structure or bedform, both in cross-section and/or plan view (as appropriate).
b. the name of the bedform/structure
c. specify the range of grain sizes in which it may be found
d. where you would look to find this bedform/structure (e.g. base of the bed, on the bedding plane, in cross-section)
e. what information it can provide you with (e.g. way up, flow depth, direction of flow, line of flow, flow strength, if the formative current is unidirectional, bidirectional or multidirectional etc)

Part 2 Identification and interpretation of flow events (The Story Tellers)
There are an additional selection of hand specimens that contain more than one sedimentary structure or bedform. These are labelled The Story Tellers.
In many, but not all, environments there are long periods of relative quiescence interrupted by brief periods of strong flow and sediment movement.

These flow events leave behind a bed or deposit as the record of the flow event. In the sedimentary record these occur as beds of relatively coarse sediment that are interbedded with relatively fine sediment or are in contact with one another. We shall call these ‘event beds’.

Some examples:

- a turbidity current deposit
- a sand bed deposited on an open marine shelf during a storm
- a bed of sand or gravel deposited on a river bed during a flood or levee or crevasse splay bed
- deposited on the floodplain

You are provided with a blank sheet of A4 paper. You should pretend this is a page in your field notebook.
Pretend that you have already made key exposure scale observations at the locality. You are now faced with making detailed qualitative and quantitative observations of **a single sandstone bed** within the interbedded sandstone and mudstone stratigraphy, so that you can interpret the palaeohydraulics of the flow at the time the event bed was deposited.

Your task is to:
(1) make a careful annotated sketch log through the specimen to summarise vertical (and where appropriate lateral) changes in grain size and sedimentary structures and bedforms

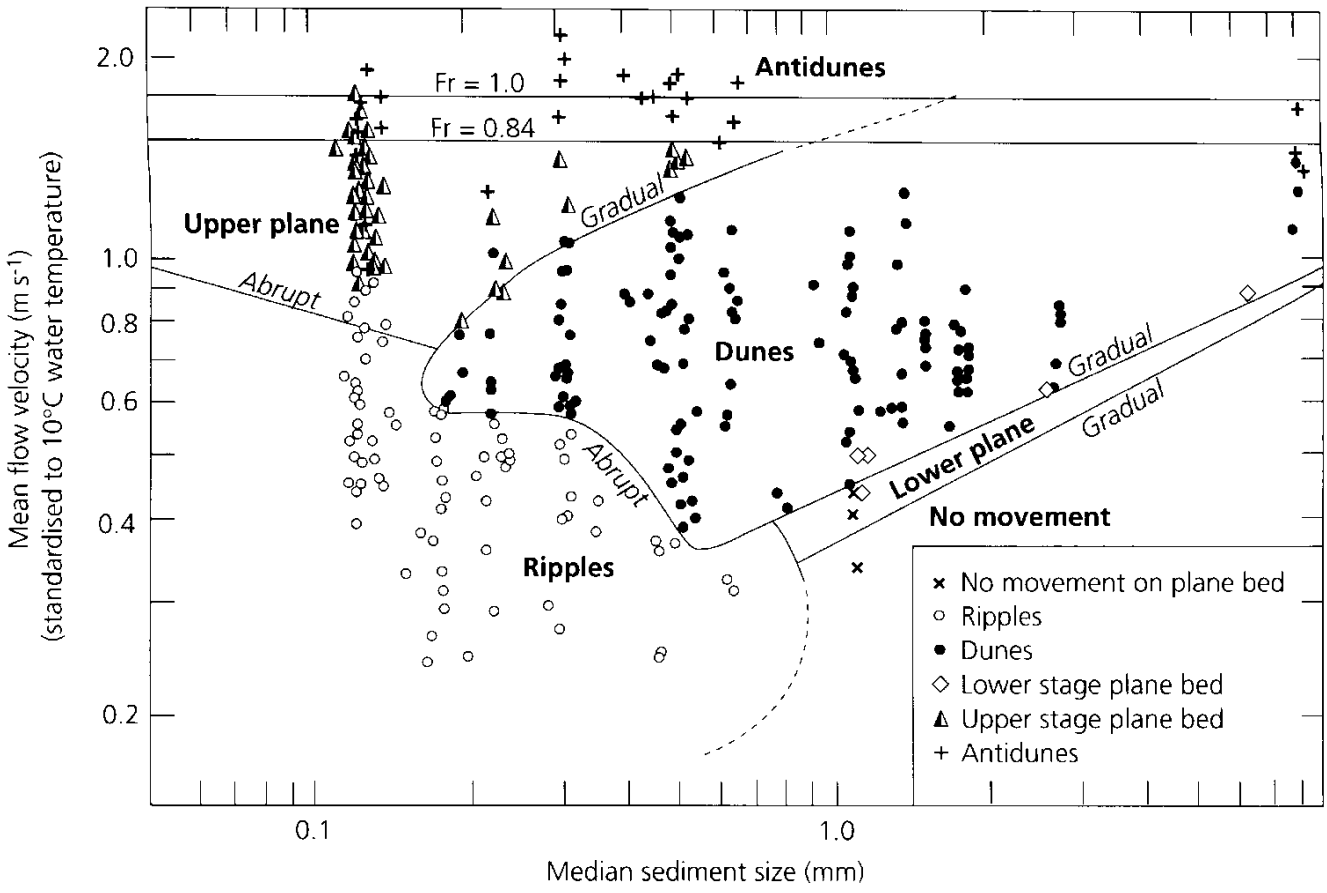
(2) specify the nature of the lower and upper contacts (e.g., loaded, erosional?)

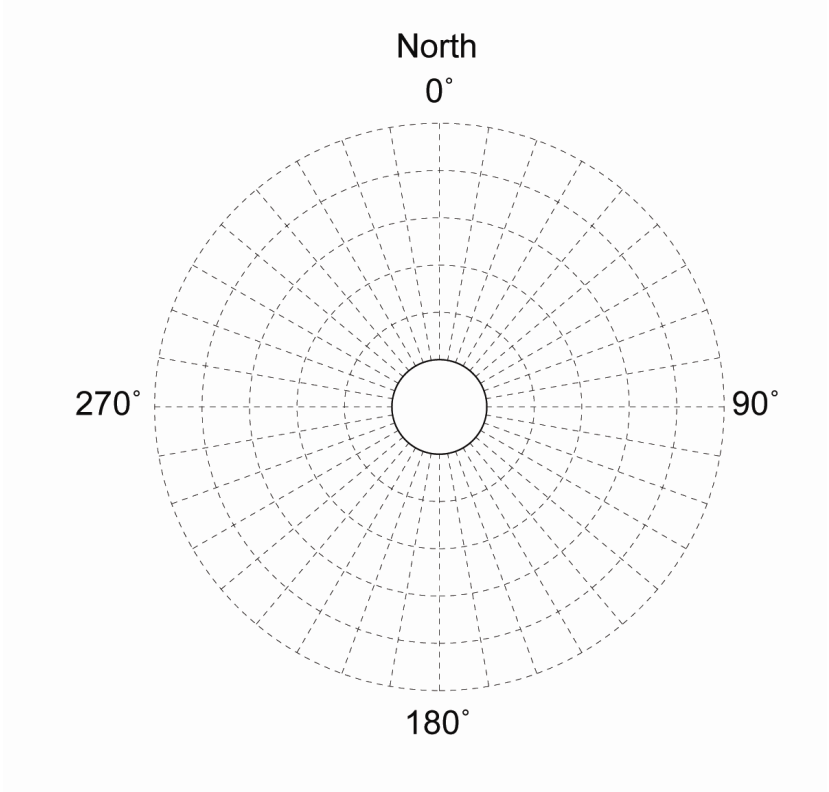
(3) interpret what you can about the flow(s) that deposited the event bed, clearly linking your interpretation to the observation that led you to that conclusion

What can you estimate about how the palaeohydraulics of the flow changed? (e.g. flow depth, strength, direction, palaeoslope etc etc).

Use bedform phase diagrams to track bedforms to provide a framework for drawing conclusions about changes in flow velocity and depth through time

- most useful for purely unidirectional and oscillatory flow, much more difficult to do for multidirectional flows
- keep in mind that there is a lag time between a change in flow and the bedform created so that the bedform may be out of equilibrium with the flow (best for ripples and high velocities, worst for dune and low velocities)


Mean flow velocity (m s-1)
(standardised to 10°C water temperature)
2.0
1.0
0.8
0.6
0.4
0.2
Fr = 1.0
Fr = 0.84
Antidunes
Upper plane
Abrupt
Gradual
Dunes
Gradual
Lower plane
Gradual
Abrupt
Ripples
No movement
× No movement on plane bed
○ Ripples
● Dunes
◇ Lower stage plane bed
◭ Upper stage plane bed
+ Antidunes
0.1
1.0
Median sediment size (mm)


North
0°
270°
90°
180°